

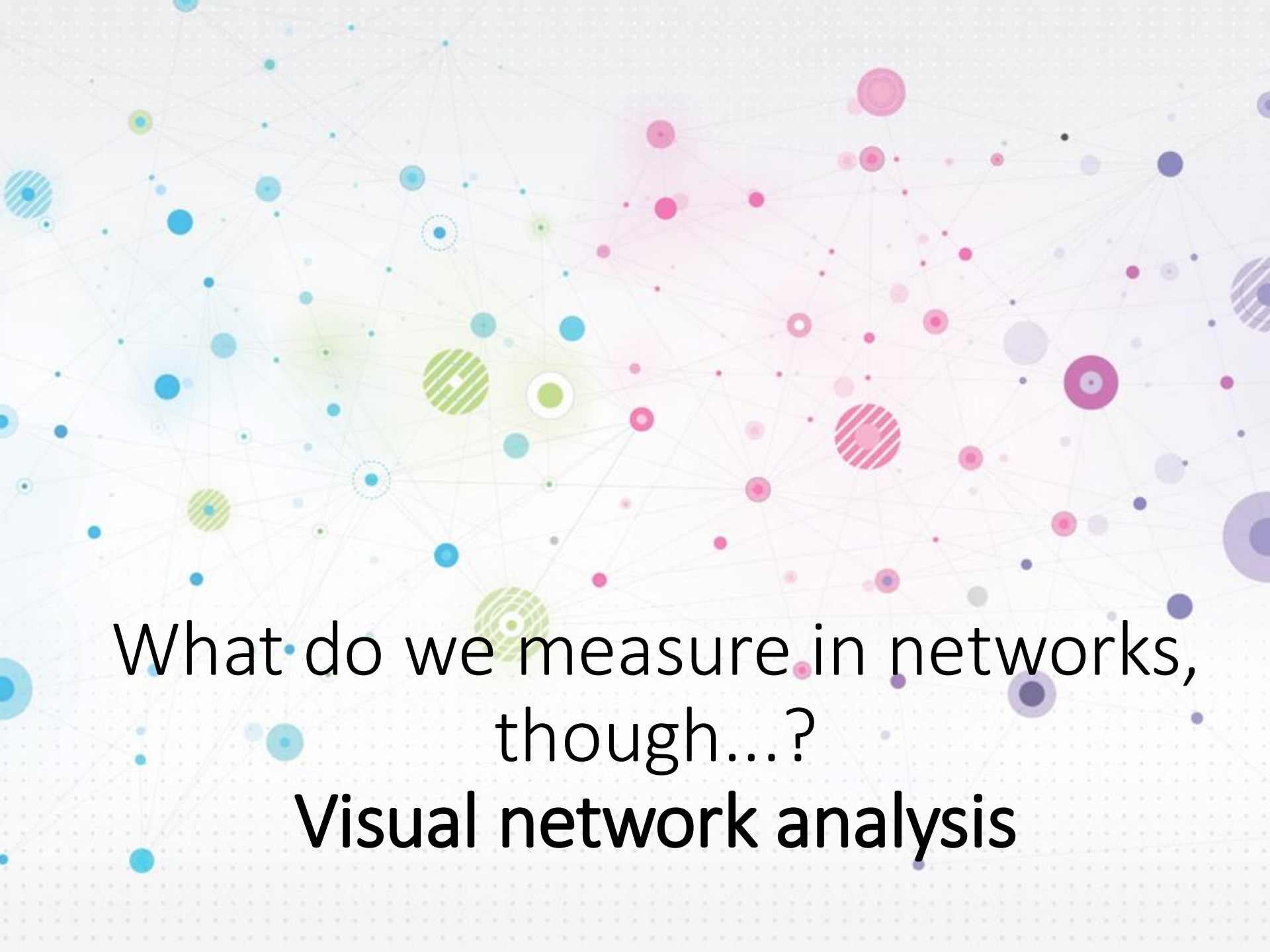
C8: Using network theories, media theories and other sociological theories in the study of social media



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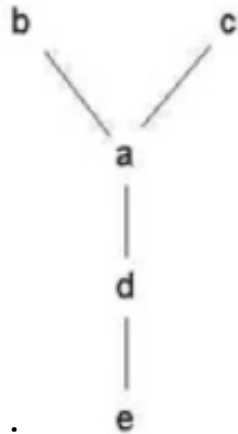
What do we measure in networks,
though...?
Visual network analysis

What is important in network graphic representation?

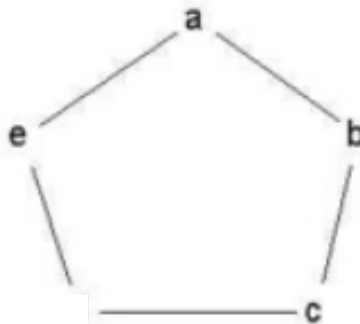
Configuration



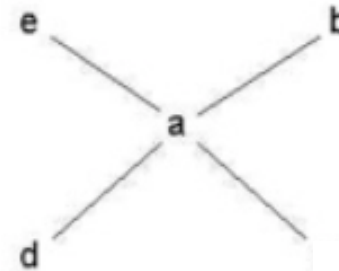
Chain
network



Y
network

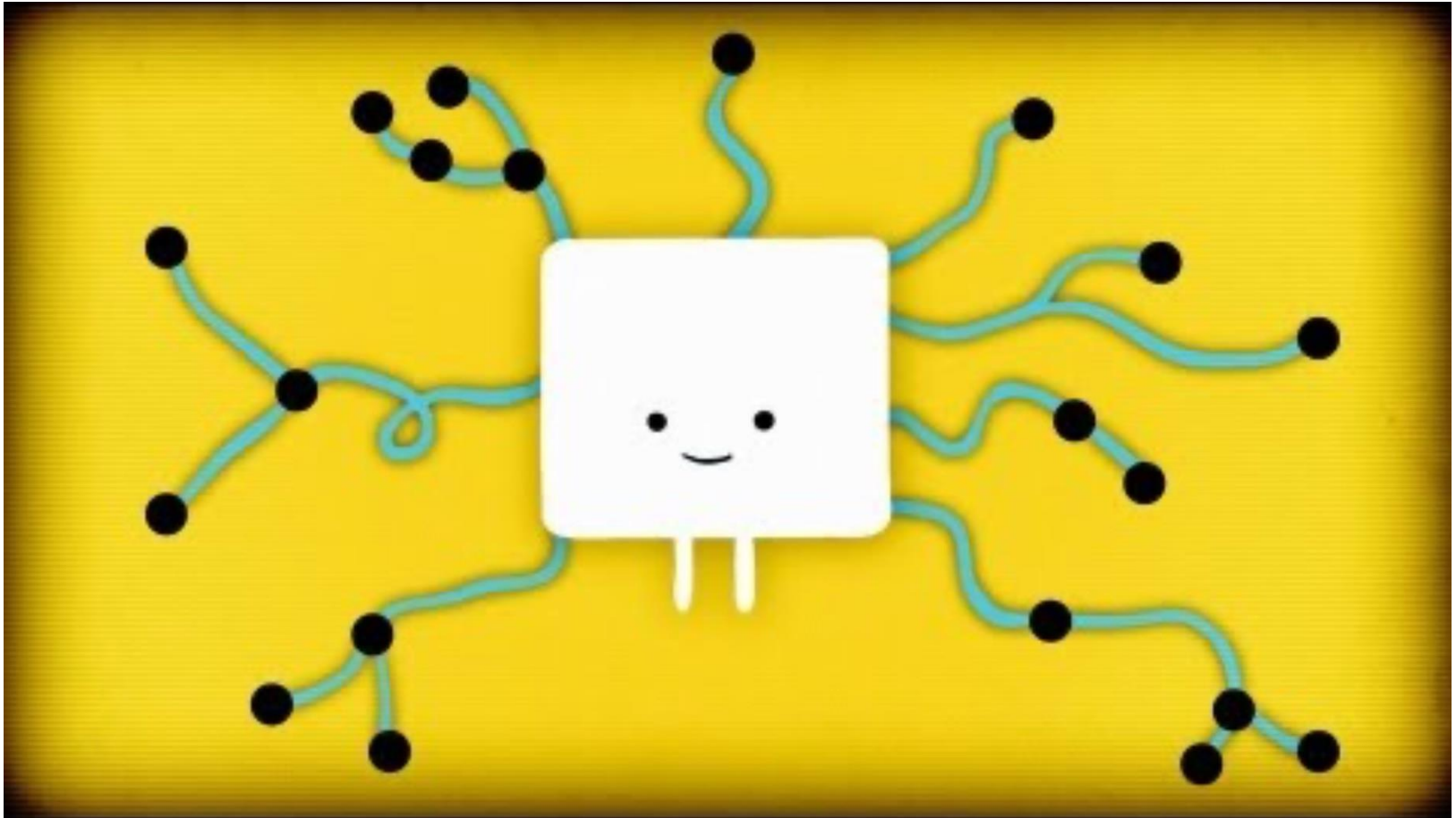


Circle
network



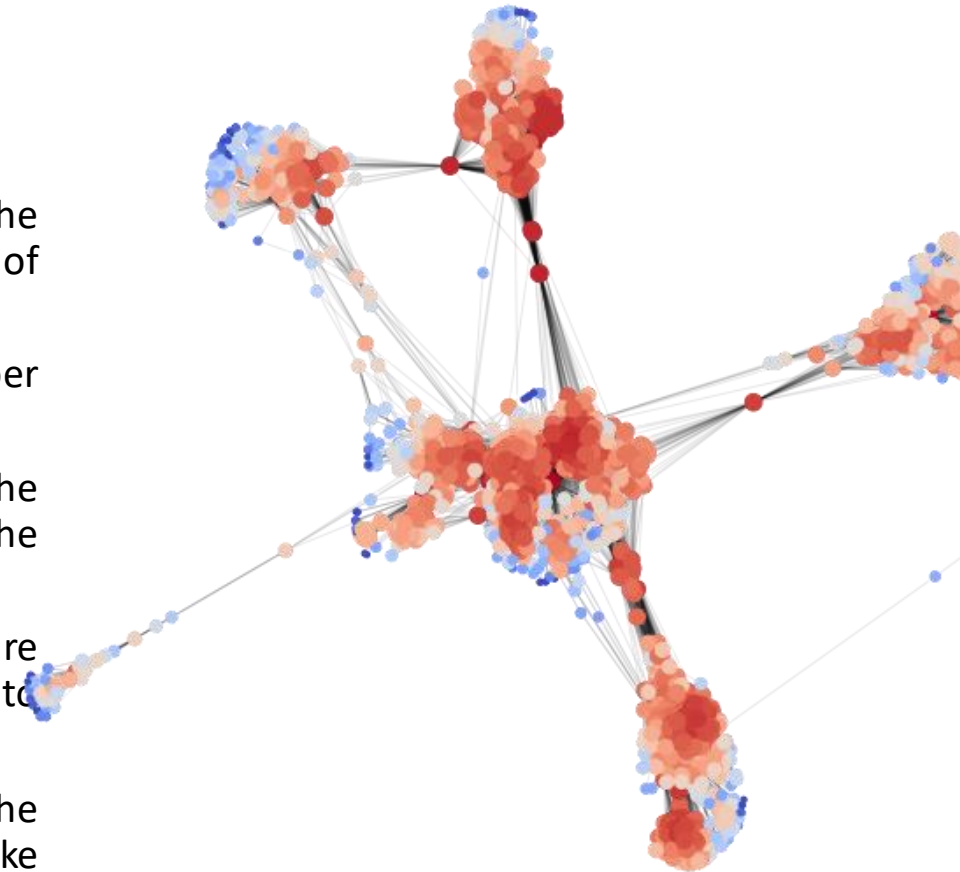
Star
network

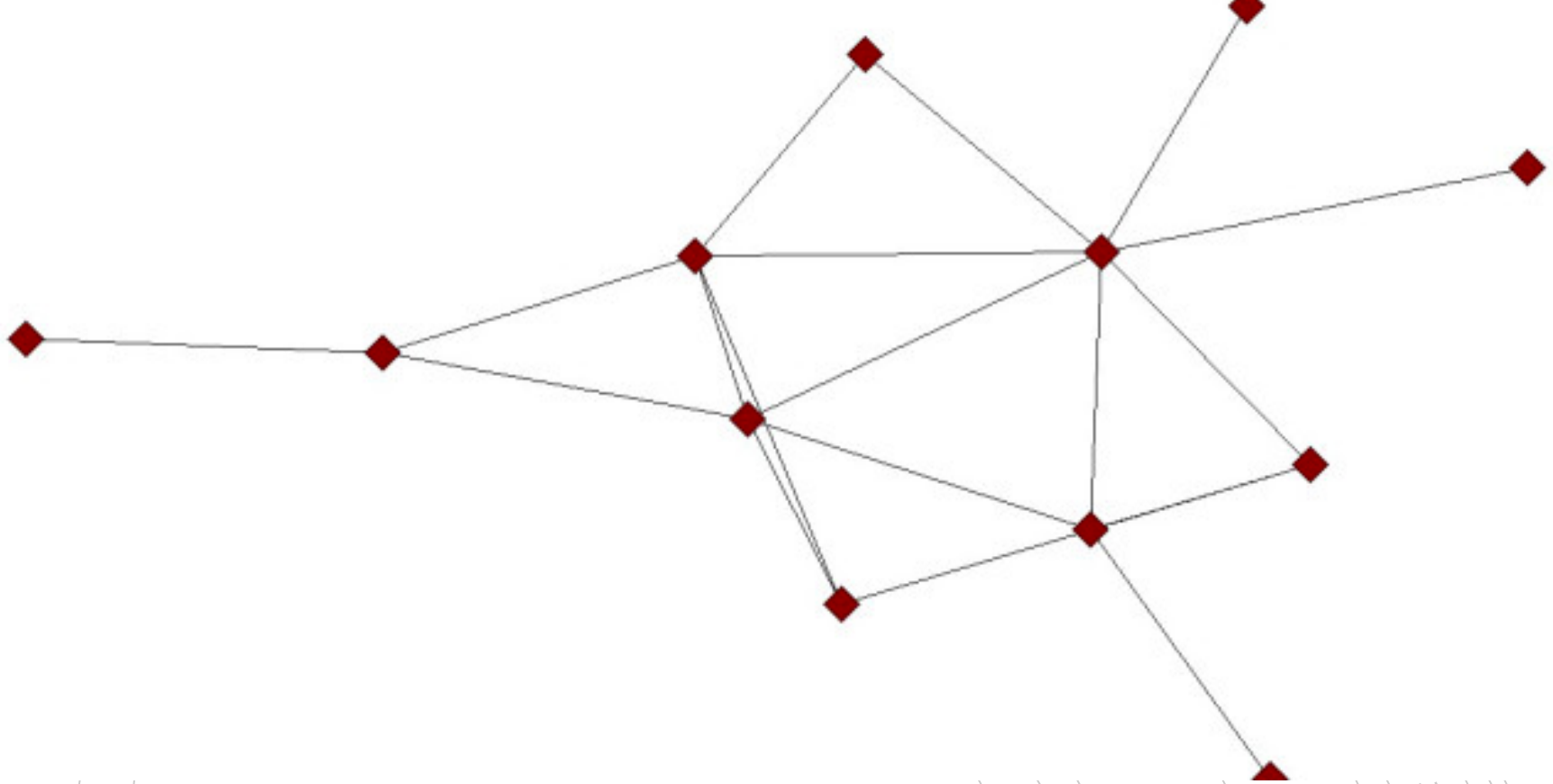
<https://youtu.be/eM1KaaTez0A?si=VTvYgiKDFUA0MSZs>



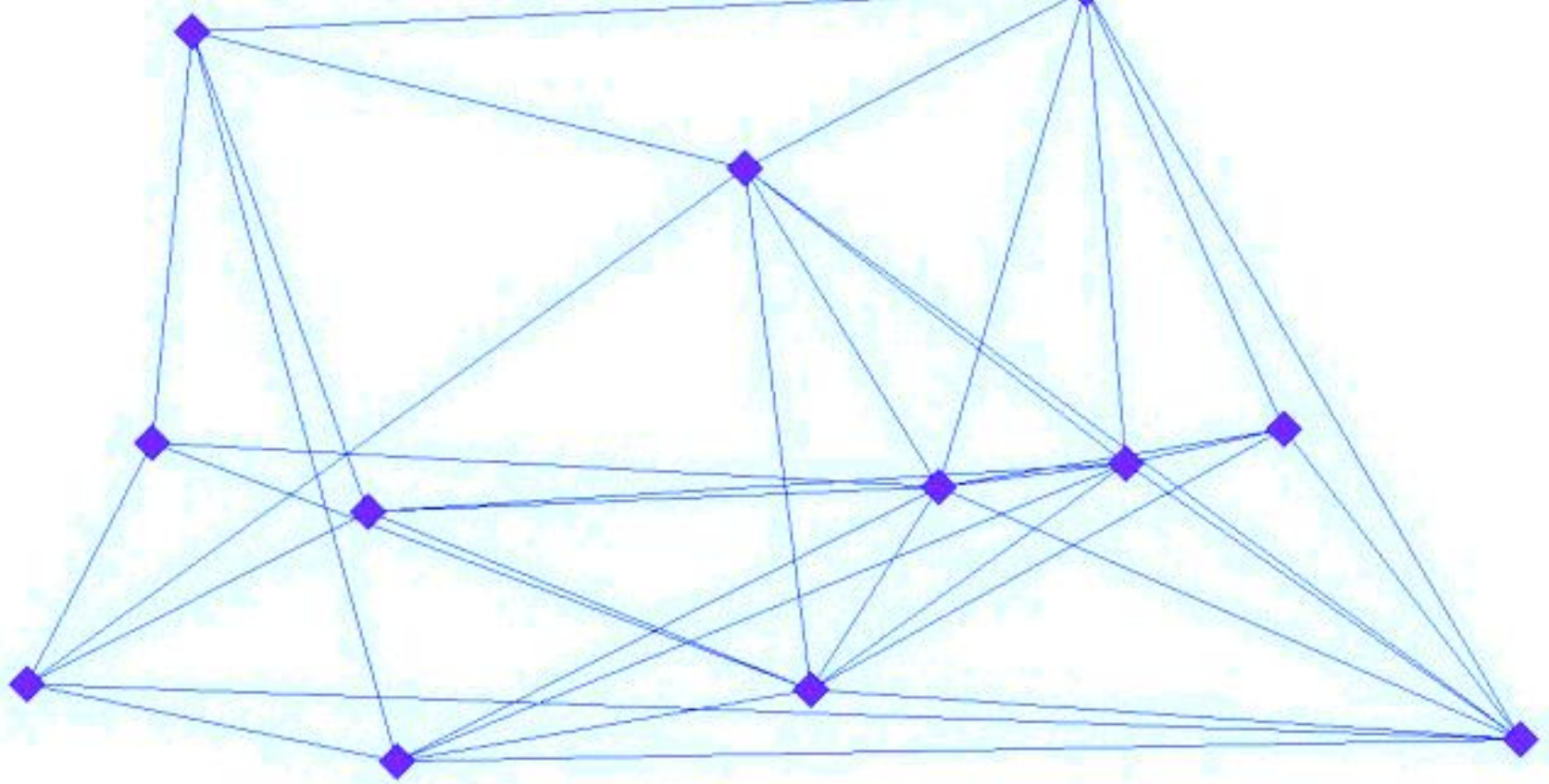
Cohesion

- **Network Density** -is the number of ties in the network expressed as a percentage of ordered/unordered pair numbers.
- **Average degree** - Average number of ties per actor.
- **Average weighted degree centrality** - is the average of the weighted average degree of the whole network.
- **Average clustering coefficient:** is a measure showing the proportion of nodes that tend to cluster together.
- **Average path length** is the average of the minimum number of steps a node needs to take to reach another node in the network. This metric is used to calculate a network's efficiency and shows the closeness of actors in a collaborative network (Ortega, 2014).





Low density

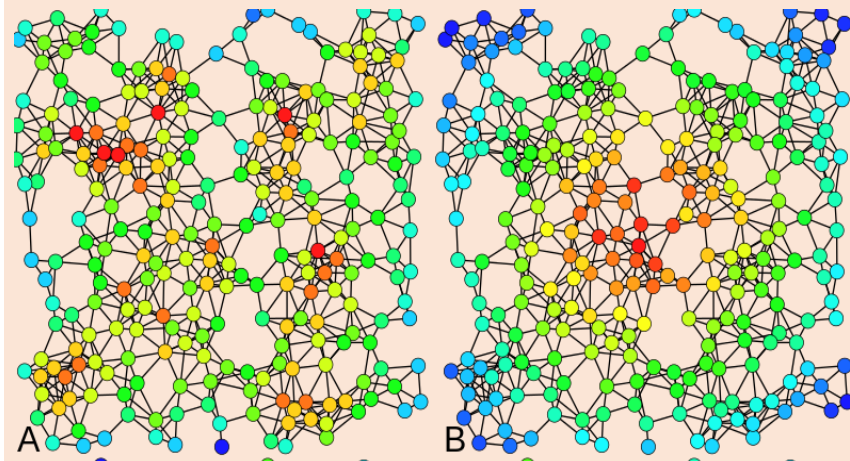


High density

The centrality

- **The centrality** is a measure of how network structure and position contribute to the importance of a node (Opsahl et al., 2010).
- The most common metrics calculated for centrality are :
- **DEGREE** – shows how well connected a particular node is, reflecting its direct influence as well as its popularity (Valente and Fujimoto, 2010). Most important are connections to actors that would otherwise remain isolated in the network. In general, most of them are connected to those in their immediate neighbourhood, to their group of friends (cliques). **A→**
 - It can be divided into two categories: **Indegree** (which is the total number of sent connections) and **Outdegree** (which is the total number of received connections).
- **CLOSENESS** measures the efficiency of access. A high value is interpreted as efficiency in communication, such as an actor's ability to disseminate and receive information quickly (Brandes et al., 2016).

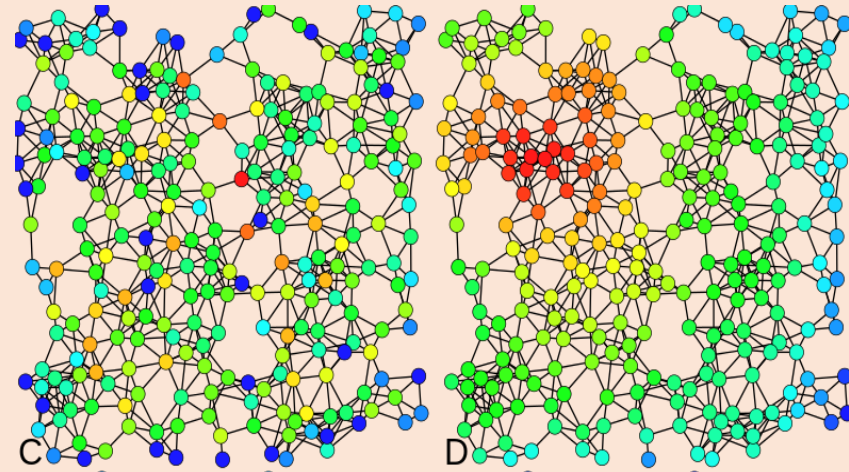
Nodes that discover the shortest paths to each other and are thus closest to any other node in the network are in a privileged position because they can monitor information flows and have an overview of what is really happening inside the network. **B→**



Source: Claudio Rocchini,
<http://www.rockini.name/math/images/centrality.svg>

The centrality

- **Connectivity** (bridges) Those who act as brokers between two important groups within a network play a powerful role, but they are also points of vulnerability that can prevent or block the flow of information.
 - **Betweenness** calculates the extent to which a node lies between another pair of nodes. Such actors represent the people who connect two otherwise unconnected subpopulations, being actors with a high input of intermediation, brokerage, surveillance and information control (Abbasi et al., 2012). **C→**
- **Eigenvector** shows the weighted graded measure in which the centrality of a node is proportional to the sum of the centralizations of the nodes to which it is adjacent → a node with a high value of this measure is connected to other highly connected nodes (representing a measure of power/influence) (Borgatti and Everett, 1997). **D→**



<https://youtu.be/RgZ1X4Dok3Y?si=Ic9Ny7GWMF1pjZCt>



Network Contagion

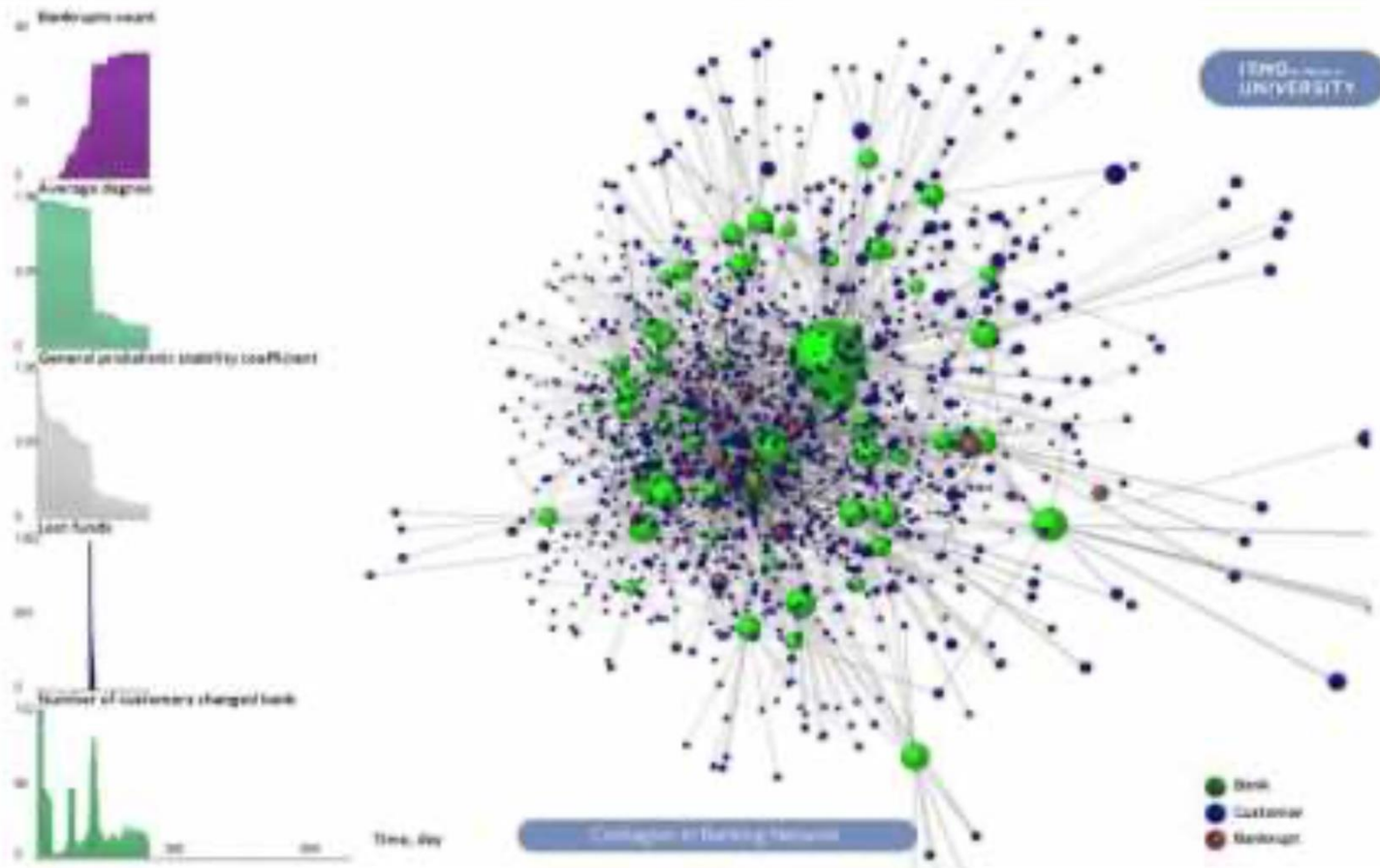
- “When you smile the whole world smiles with you”

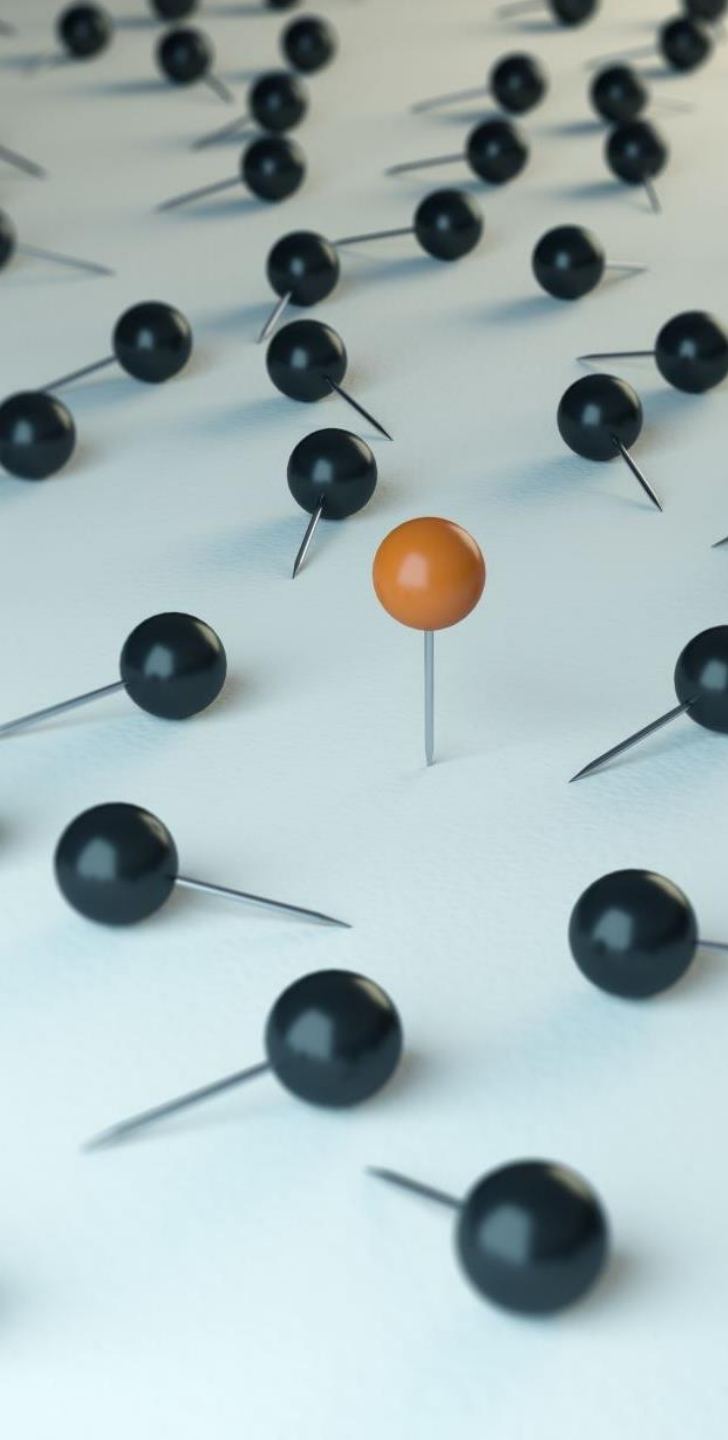
Christakis, Nicholas & Fowler, James (2009)
–Chapter 2

Mandatory readings c7



<https://youtu.be/FpMKi9gZJrY?si=vGU9xzaKgkGRUFy5>





Network communication

- Zhang, S. S., Liang, X., Wei, Y. D., & Zhang, X. (2020). On structural features, user social behavior, and kinship discrimination in communication social networks. *IEEE Transactions on Computational Social Systems*, 7(2), 425-436.

<https://ieeexplore.ieee.org/document/8954613>

Watch the presentation of Cesar Hidalgo - Meritocracy and Topocracy of Networks (10:58 min) - the dispersion and structure of networks is a fundamental constraint on the meritocracy of societies: <https://youtu.be/CTJ8TAMv3sk>



Social capital

- Xu, W., & Saxton, G. D. (2019). Does stakeholder engagement pay off on social media? A social capital perspective. *Nonprofit and Voluntary Sector Quarterly*, 48(1), 28-49.
<https://doi.org/10.1177/0899764018791267>
- STUDY To what extent do social work graduates have the digital skills needed to connect with their peers in a strategic way, thus enabling them to gain sufficient social capital, self-esteem, social satisfaction and well-being? – „Social Networking Sites and Youth Transition: The Use of Facebook and Personal Well-Being of Social Work Young Graduates” -
<https://www.frontiersin.org/articles/10.3389/fpsyg.2020.00230/full>



Bourdieu (1984): Theory of Social Capital

- Social capital is understood as a set of resources related to the possession of a network of stable relationships of mutual knowledge and recognition - or, in other words, belonging to a group.
- provides each of its members with the support of collectively held capital,
- The benefits of belonging to a group form the basis of solidarity.
- all our actions are influenced by contextual factors (habitus), such as our level of education or social origin, even by the criteria for appreciating something (tastes)
- **See** Williams, J. R. (2019). The use of online social networking sites to nurture and cultivate bonding social capital: A systematic review of the literature from 1997 to 2018. *New Media & Society*, 21(11-12), 2710-2729 - https://journals.sagepub.com/doi/pdf/10.1177/1461444819858749?casa_token=zGsuV0n5C58AAAAA:03_KqB9ptcj2HqcgAols8tV2q_e92oVT96iRyPE0Cba6WZbXYtn61WY8fDNLQD3uj0QWqgi ohR2mw



What do we
measure in
media
research

- Consumer behaviour
- Effects of media consumption
- Reasons for consumption

Social media consumption

Kitchens, B., Johnson, S. L., & Gray, P. (2020). Understanding Echo Chambers and Filter Bubbles: The Impact of Social Media on Diversification and Partisan Shifts in News Consumption. MIS Quarterly, 44(4): https://www.darden.virginia.edu/sites/default/files/inline-files/05_16371_RA_KitchensJohnsonGray%20Final_0.pdf

Heffner, J., Vives, M. L., & FeldmanHall, O. (2021). Anxiety, gender, and social media consumption predict COVID-19 emotional distress. Humanities and Social Sciences Communications, 8(1), 1-7 <https://www.nature.com/articles/s41599-021-00816-8.pdf?origin=ppub>

Auxier, B., & Anderson, M. (2021). Social media use in 2021. Pew Research Center, 1, 1-4. https://www.pewresearch.org/inter-net/wp-content/uploads/sites/9/2021/04/PI_2021.04.07_Social-Media-Use_FINAL.pdf

Ortiz-Ospina, E., & Roser, M. (2023). The rise of social media. Our world in data. <https://ourworldindata.org/rise-of-social-media?ref=tms>

<https://youtu.be/HVQGYANFRsc?si=g38qzAqliHbSbEC->



<https://youtu.be/4TMPXK9tw5U?si=tBAU2myh0okU3SnS>





Social media effects

- Orben, A. (2020). Teenagers, screens and social media: a narrative review of reviews and key studies. *Social psychiatry and psychiatric epidemiology*, 55(4), 407-414.
https://link.springer.com/article/10.1007/s00127-019-01825-4?wt_mc=Internal.Event.1.SEM.ArticleAuthorOnlineFirst&utm_source=ArticleAuthorOnlineFirst&utm_m...
- Orben, A., Dienlin, T., & Przybylski, A. K. (2019). Social media's enduring effect on adolescent life satisfaction. *Proceedings of the National Academy of Sciences*, 116(21), 10226-10228.
<https://www.pnas.org/doi/abs/10.1073/pnas.1902058116>
- Hogue, J. V., & Mills, J. S. (2019). The effects of active social media engagement with peers on body image in young women. *Body image*, 28, 1-5.
<https://www.sciencedirect.com/science/article/pii/S174014451730517X>

Reasons (motivations) for using social media

- Hossain, M. A. (2019). Effects of uses and gratifications on social media use: The Facebook case with multiple mediator analysis. *PSU Research Review*.
<https://www.emerald.com/insight/content/doi/10.1108/PRR-07-2018-0023/full/html>
- Pelletier, M. J., Krallman, A., Adams, F. G., & Hancock, T. (2020). One size doesn't fit all: a uses and gratifications analysis of social media platforms. *Journal of Research in Interactive Marketing*, 14(2), 269-284.
https://www.researchgate.net/profile/Mark-Pelletier/publication/341833793_One_size_doesn't_fit_all_a_uses_and_gratifications_analysis_of_social_media_platforms/links/5f035f81a6fdc4ca44ebf7e/One-size-doesnt-fit-all-a-uses-and-gratifications-analysis-of-so
- Sun, Y., Liu, Y., & Zhang, J. (2020). Excessive enterprise social media use behavior at work: role of communication visibility and perspective of uses and gratifications theory. *IEEE Access*, 8, 190989-191004.
<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9230429>
- Chugh, R., & Ruhi, U. (2018). Social media in higher education: A literature review of Facebook. *Education and Information Technologies*, 23(2), 605-616.
<https://link.springer.com/article/10.1007/s10639-017-9621-2>





Other social theories
used in the study of
social media

Goffman (1959/ 2003) The presentation of self in everyday life

- Self-presentation: When an individual appears in front of others, there is usually a reason for him to mobilise his activity so as to convey to others an impression that is in his best interest.
- Appearances - impression management: how the individual, in ordinary situations, presents himself and his activity to others, how he guides and controls the impression they form of him, and the kinds of things he may or may not do while sustaining his performance in front of them
- See: Murwani, E. (2018). The Impression Management Strategy of the Candidates of Governor-Vice Governor of DKI Jakarta on Social Media. *Jurnal Komunikasi Ikatan Sarjana Komunikasi Indonesia*, 3(2), 113-121:
<http://download.garuda.kemdikbud.go.id/article.php?article=876071&val=10680&title=The%20Impression%20Management%20Strategy%20of%20the%20Candidates%20of%20Governor-Vice%20Governor%20of%20DKI%20Jakarta%20on%20Social%20Media>
- Also, see: Merunková, L., & Slerka, J. (2019). Goffman's theory as a framework for analysis of self presentation on online social networks. *Masaryk UJL & Tech.*, 13, 243.
<https://journals.muni.cz/mujlt/article/download/11836/10855>



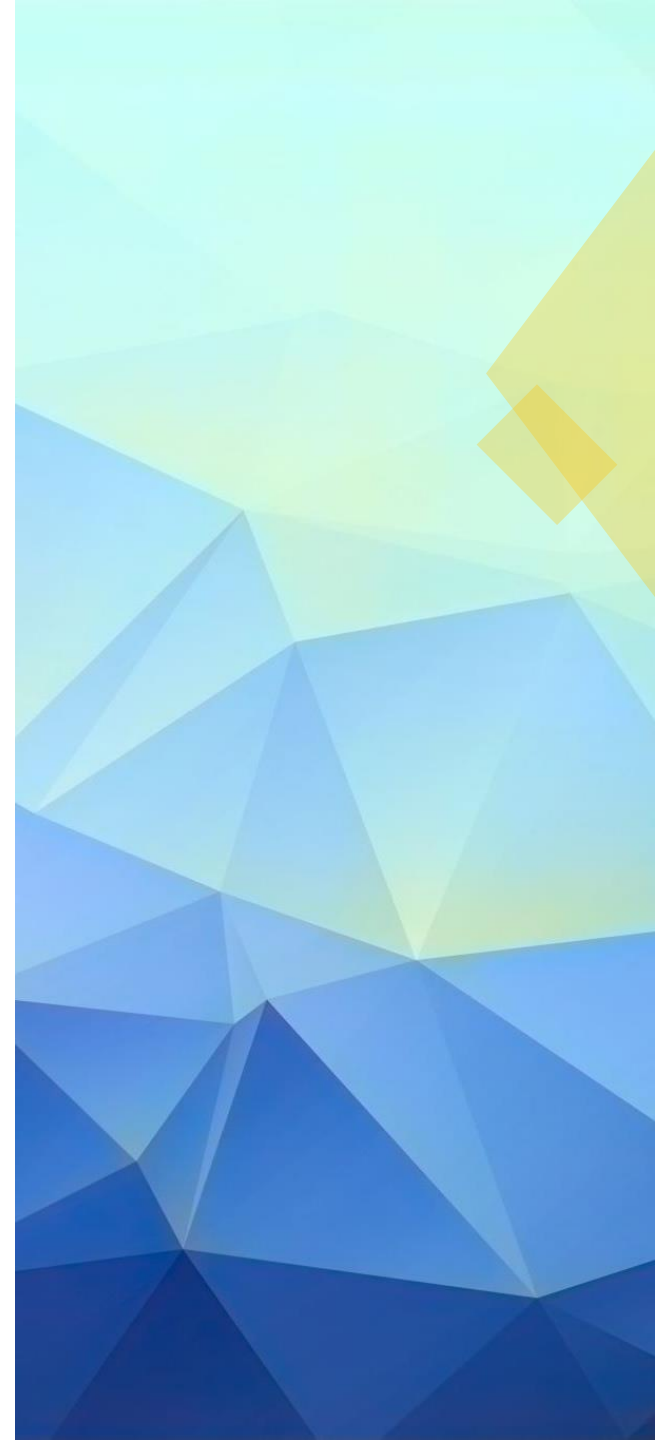
Bandura (1977, 1989): Social learning theory

Individuals' behaviours are socially acquired by observing and interacting with others.

Learning is the result of a reciprocal interaction between the person, the external social context and the behaviours adopted to achieve goals (responses to stimuli);

See Deaton, S. (2015). Social learning theory in the age of social media: Implications for educational practitioners. *Journal of Educational Technology*, 12(1), 1-6.

<https://files.eric.ed.gov/fulltext/EJ1098574.pdf>



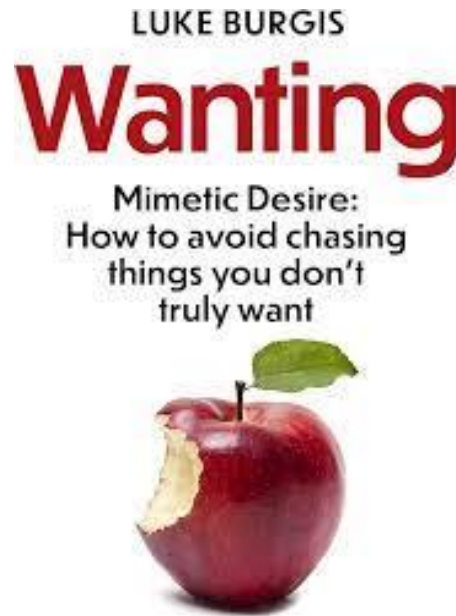
Kelman (1959): social influence theory

Individual knowledge and behaviour can be affected by three social processes: conformity (normative influence of others' expectations), internalisation (congruence of one's own goals with those of other group members) and identification (self-concept in terms of the defining characteristics of the group).

They shape an individual's social identity;

See: Liu, Y., Chen, Y., & Fan, Z. P. (2021). Do social network crowds help fundraising campaigns? Effects of social influence on crowdfunding performance. *Journal of Business Research*, 122, 97-108.
https://www.sciencedirect.com/science/article/pii/S0148296320305506?casa_token=hwH4uMynrWAAAAA:dNXu8P9u57-WjVd2DQI9kJqaC9Ryg9N0dkbgrUze9Jn18zdJ0lIgRQU5QNGqfwWfUv0jXuBZqE

Burgis (2021): *Wanting: The Power of Mimetic Desire in Everyday Life*



- insights from philosophy (Thomas Aquinas, Sartre, Rene Girard) and neuroscience;
- Are we autonomous? Do we make only rational choices? Do our desires come from deep within? Are we above the influence?
- The models we encounter in our daily lives, be they celebrities, influencers, or even our peers, exert a profound influence on our desires. These are what we refer to as 'mimetic models'.
- Ecology of desire
- Mimetic desire typically works beneath the surface of our conscious awareness, influencing our decisions without fully recognizing its impact.
- Effects: It influences our feelings of self-worth and desires and catalyses cultural and political polarization.
- In the time of hyperimitation → social media has fueled mimesis because it encourages imitation by its very structure.
- Empathy to counteract mimesis

<https://youtu.be/dtBtov2f7e4?si=-OlPGMHXLgbfwrO>



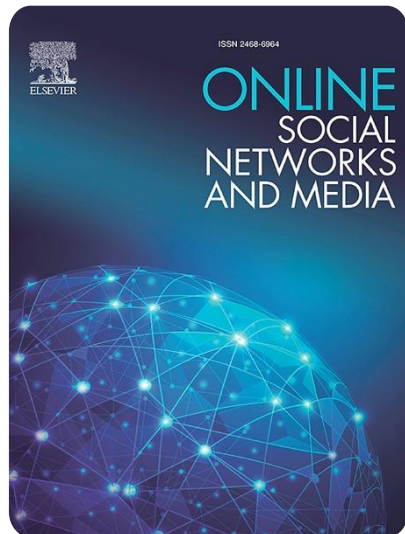
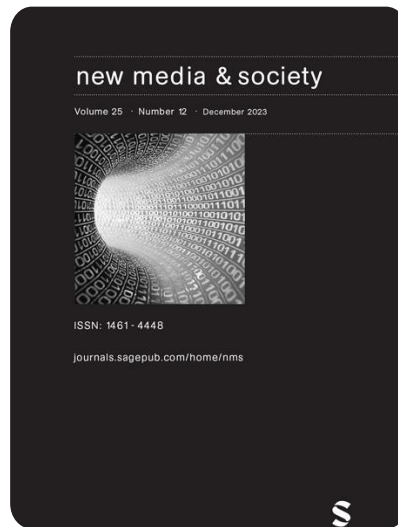
<https://youtu.be/Dclgk94Fp6Y?si=0vLWaWv2cEXsTNUf>



SM+S
social media
+ society



O



Academic journals where we find studies about social media

- **Social Media + Society** is a peer-reviewed, open access journal that focuses on promoting understanding of social media and its impact on societies past, present and future.
<https://journals.sagepub.com/home/sms>
- **New Media & Society** is a journal that addresses critical discussions of key issues arising from the scale and speed of new media development, drawing on a wide range of disciplinary perspectives and theoretical and empirical research.
<https://journals.sagepub.com/home/nms>
- **Online Social Networks and Media** is an international peer-reviewed journal that publishes both theoretical and experimental scientific articles and research papers covering all aspects of online media and social networks: from protocols and applications to the use of data extracted from these online networks for modelling and understanding human behaviour in the cyber-physical world.
<https://www.sciencedirect.com/journal/online-social-networks-and-media>



Academic journals where we find studies about social media

- **Media and Communication** is an open - access, peer-reviewed, international journal devoted to a wide range of fundamental and applied research in communication and related fields.
<https://www.cogitatiopress.com/mediaandcommunication>



- **Journal of Computer-Mediated Communication** is an open - access scientific journal. It focuses on social science research on communication using computer-based media technologies. Within this general field, the journal is broadly interdisciplinary, publishing work by researchers in communication, business, education, political science, sociology, psychology, media studies, information science and other disciplines
<https://academic.oup.com/jcmc>

Reading scientific articles (Literature review)



Title, Author(s), Year of Publication



Keywords



Background of the study (premises)



Research objectives (Research questions)



Literature - concepts and main theories underlying the scholarly approach



The methodology used (Type, methods and procedure)



Findings (results)



Discussion and conclusions